

Inter-Subject Correlation of Heart Rate and Attention Modulation

Roberto Antonello 47891A

Abstract

This project focuses on analyzing heart-rate synchrony across individuals to verify if it's modulated by attentional state while watching educational videos. The goal is therefore to replicate a couple of results from Pérez et al. (2021) [1] using data from a newer dataset: the Brain, Body and Behavior Dataset (BBBD) [2].

To achieve this, heart-rate fluctuations were extracted from raw ECG recordings and compared across subjects using ISC-HR (inter-subject correlation of heart-rate).

A hypothesis test with a FDR correction is used to establish statistical significance of the obtained results. ISC-HR is then compared between attentive and distracted conditions through a paired statistical test across all five stimuli.

Hypothesis test and descriptive statistics are employed to analyze synchrony patterns and to detect the effect of attention across stimuli.

The results provide a statistically grounded replication of ISC-HR modulation by attention, discussed in relation to the original paper's finding on which this project is based.

1 Introduction

Heart rate is often treated as a purely autonomic signal, reflecting internal physiological states such as arousal or stress.

Recent studies suggest that heart-rate fluctuations can be driven by external stimuli. When multiple individuals are exposed to the same narrative content, their heart rates tend to fluctuate in a similar way.

This phenomenon is known as inter-subject correlation of heart-rate (ISC-HR).

Pérez et al. (2021) [1] proposed that these heart-rate fluctuations are modulated by conscious, attentive processing of the stimulus.

This project aims to replicate two of the main results reported by the analyzed paper.

- ISC-HR is statistically significant across subjects exposed to the same stimulus (R1).
- Synchrony is modulated by attentional state, being higher when subjects are attentive than when they are distracted (R2).

This work uses data from the Brain, Body and Behavior Dataset (BBBD) [2], which is a more recent dataset that includes similar experiments to the original study.

2 Related Work

here put info about article on we based the analysis.

Il paper di Pérez et al. (2021) — sintesi dell'ipotesi centrale e dei risultati principali su cui ti basi.

3 Dataset

here put info about dataset used

BBBD, Esperimento 2 — struttura (36 soggetti nominali, 5 video, 2 condizioni), e la nota sulla discrepanza temporale (paper 2021 vs dataset 2025) che abbiamo appena scritto — questa sezione può attingere quasi di peso dal box che ho aggiunto in Final-Results-R1-R2-Discussion.pdf.

4 Methodology

introduction to methods used Qui hai già quasi tutto pronto, solo da sintetizzare (i documenti supplementari sono la versione "lunga", qui ne fai il riassunto):

4.1 Signal preprocessing

da HR-Pipeline-Note.pdf + Butterworth-Filter-Note.pdf: filtro, picchi, HR istantanea, interpolazione

4.2 ISC-HR computation

da Hypothesis-Test-R1-Note.pdf Sezione 1: Fisher-Z, Step 3-5 del paper

4.3 Hypothesis test

permutazione circolare, Benjamini-Hochberg

4.4 Attention modulation

da R2-Attention-Modulation-Note.pdf: correlazione a riferimento fisso, esclusione dell'auto-correlazione, Shapiro-Wilk + t-test/Wilcoxon.

4.5 Implementation choices

(opzionale, se vuoi una tabella riassuntiva) — QC, i parametri non specificati dal paper, tutti già tracciati nei documenti

5 Results

Qui attingi principalmente da Final-Results-R1-R2-Discussion.pdf, che è già scritto quasi in forma di risultati/discussione:

5.1 Input validation

5.2 R1 on all combinations of (session, stimulus)

5.3 Significant stimulus

5.4 Stimulus 02 case

6 Discussion

Confronto con le cifre del paper (con le cautele che abbiamo già scritto), limiti dichiarati (niente ANOVA completa, sensibilità all'interpolazione non testata, scelte QC/esclusione non specificate dal paper), interpretazione del pattern generale.

7 Conclusion

Cosa hai dimostrato, cosa resta aperto (R3, metodo esatto per test multipli, ecc.).

8. References

Paper Pérez et al., documentazione BBBD/Zenodo, eventuali librerie citate.

Declarations

Ai Usage Disclaimer: parts of this project have been realized with the assistance of ChatGPT(model GPT-5.3) and Claude(model Sonnet 4.6) for the following types of tasks:

- Reviewing and refining text during the writing of this report.
- Collecting ideas about results analysis.
- Generation of Python code to support the development.

Every content generated by AI has been reviewed and tested by the author.

References

- [1] P. Pérez, J. Madsen, L. Banellis, B. Türker, F. Raimondo, V. Perlberg, M. Valente, M.-C. Niérat, L. Puybasset, L. Naccache, T. Similowski, D. Cruse, L. C. Parra, and J. D. Sitt. Conscious processing of narrative stimuli synchronizes heart rate between individuals. *Cell Reports*, 36(11):109692, 2021.

- [2] J. Madsen, N. Kuppa, and L. C. Parra. The Brain, Body, and Behavior Dataset (BBBD): Multimodal Recordings during Educational Videos. *Scientific Data*, 13:920, 2026.